Lesson 2: Key Press Reactor

Scratch

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Open Scratch (browser or desktop). Start a new project if one isn't already open.
2. Save your project as [YourName]Game - this is the start of your Semester 1 project.
3. Keep the default cat sprite, or import one of the placeholder images from this lesson's Assets folder (Upload Sprite > choose player_circle_blue.png or similar).

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

4. Click the Events category (yellow blocks).
5. Drag "when [right arrow] key pressed" into the code area.
6. Click the Motion category, drag "change x by 10" underneath it.
7. Click the stage, then press the right arrow key to test.

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- Repeat with 4 separate scripts - "when [up/down/left/right] arrow key pressed" - each using a Motion block ("change y by 10", "change y by -10", "change x by -10", "change x by 10") so your sprite moves in all 4 directions.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: make your sprite "blink" while moving and stay solid when idle. This needs its own "when green flag clicked" script with a forever loop, checking the Sensing category's "key [ ] pressed?" reporter for all 4 arrow keys - that's the one part of this exercise that genuinely needs a loop, because Scratch has no "when key released" block to trigger off of.

- Sound: play sfx_blip.wav once each time you press a movement key - silent the rest of the time, since a "when key pressed" script only runs at the instant of the press. Upload it from this lesson's Assets folder (Choose a Sound > Upload Sound) and add a "start sound" block right inside each of your 4 movement scripts - no loop or timer needed.
- Recolor: use a different costume for each direction, switching back to your default costume when idle - swap the picture rather than tint it. Upload the other placeholder circles from this lesson's Assets folder as extra Costumes on your sprite (Costumes tab > Upload Costume): red, blue, green, and yellow for the 4 directions, plus orange as your dedicated idle costume even if you started with a different one in Setup. Switch costume the moment you press a key (inside that key's own script), and switch back to orange in the same green-flag loop you built for the blink - since "returning to idle" needs to know when no key is held anymore, same as blink does.
- Bounce: add "if on edge, bounce" (Motion) right after the movement block inside each of your 4 key-press scripts, so your sprite bounces off the stage edge instead of walking off it. Heads up: this can flip your sprite left-right when it bounces, depending on its rotation style - that's a known Scratch quirk, not a bug, and it's fine to leave as-is for now. (Precisely stopping at the edge with no flip needs the position-checking technique from a later lesson.)
- Structure: keep this as 5 separate scripts, not one - "when green flag clicked" for the blink/idle-reset loop, plus one "when key pressed" script per arrow key that handles that direction's movement, sound, color, and bounce together. Each key-press script IS the "if this key, do this" logic - you don't need a separate "if key pressed?" check inside it. See the Key for the fully worked version.

If Something's Wrong

- Nothing happens when you press a key: make sure the blocks are actually snapped together (a white highlight shows when they connect), and click the stage first so Scratch is "listening."
- Sprite only moves once even holding the key: that's normal - each press triggers one run. Continuous movement while held is a Level 3 stretch goal using a "forever" loop with a "key [ ] pressed?" check inside an if-block.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas